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Named Entity Recognition in German-language Telegram channels

Bachelor Thesis

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by

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Abstract

This thesis implements Named Entity Recognition for German-language text messages from the messenger service Telegram, focussing on a Telegram channel mobilizing for protests against COVID-19 containment measures in Germany.

The primary goal is to explore the potentials and limitations of two existing German-language models for Named Entity Recognition, namely *spacy* and *flair*, for the detection of entities of the common entity types PERSON, LOCATION, and ORGANIZATION.

Furthermore, the recognition of an additional custom entity type ACTION will be explored with the help of manually labeled data used for training custom models. Text parts containing DATE and TIME information are extracted via the *ctparse* module, and

This thesis contributes to the task of German-language NER in general by applying it to a specific domain and data type, namely text data from messenger services. Furthermore, its aim is to provide a tool for both researchers and civil society for monitoring and analyzing large amount of text data from social media/messengers.

Dedication

Statutory Declaration

I herewith formally declare that I have written the submitted thesis independently. I did not use any outside support except for the quoted literature and other sources mentioned in the paper.
I clearly marked and separately listed all of the literature and all of the other sources which I employed when producing this academic work, either literally or in content.
I am aware that the violation of this regulation will lead to the failure of the thesis.

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Chapter 1

Introduction

1.1 Background

The outbreak of the COVID-19 pandemic was followed by an increasing spread of misinformation and conspiracy theories. Those narratives evolved not only around the virus itself, but also about the intentions of those responsible for putting into effect measures for its containment.

Those measures included the mandatory use of face masks, rapid testing, and vaccinations, at times functioning as admission requirements in the workplace or publicly accessible spaces such as restaurants and theatres. Furthermore, unprecedented restrictions were imposed on public life through several general or partial lock-downs of social and economic life, as well as temporary bans of gathering in public. These restrictions also heavily affected the possibilities to hold political demonstrations.

From the beginning on, these measures were accompanied by critique and protests from various political spectres. In Germany, the first noticeable protests against COVID-19 measures took place at the so-called 'Hygienedemonstrationen' (hygiene demonstrations) in front of the 'Volksbühne' theatre in Berlin. Over time, the so-called 'Querdenken' (lateral thinking) movement became the most prominent movement evolving around protests against the COVID-19 containment measures. The movement reached its preliminary climax with several large demonstrations taking place in Berlin, directed against legislative changes regarding the containment of the pandemic passed by the German parliament. In the context of these demonstrations, violent clashes occurred between police and protesters, which often refused to act according the official restrictions for the demonstration such as wearing a face mask. Furthermore, there were several incidents in which journalists were violently attacked by protesters, accusing them to be part of the 'Lügenpresse' (lying press), and two incidents in which the seat of the German parliament, the Reichstag, was targeted by protesters: On August 29th 2020, around 500 protesters, among them several prominent actors from the German far-right movement, broke the barriers in front of the Reichstag in Berlin, seat of the German parliament, and occupied the stairs of the building, waving several 'Reichskriegsflaggen'. On November 18th 2020, another large demonstration organized by the 'Querdenken' movement took place; once again accompanied by heavy clashes between protesters and police officers. On the same day, a small group of right-wing social media activists was invited as visitors to the Reichstag by the German far-right party 'Alternative für Deutschland' (AfD). The group approached and harassed politicians from other parties, filming and live-streaming the events, an incident which was perceived as a caesura and heavily criticized by the other parties.

On April 21st 2021, the last large authorized demonstration of the 'Querdenken' movement took place in Berlin, comparing a new law for the containment of the pandemic with the 'Ermächtigungsgesetz' put into effect in 1933 which ultimately paved the way for the Nazi dictatorship in Germany. In response to several violations of official restrictions by demonstration participants, police decided to disperse the demonstration. Several protesters subsequently moved to other areas nearby. Clashes between protesters and police continued to occur throughout the rest of the day.

The subsequent demonstrations announced by the 'Querdenken' movement to take place in Berlin were prohibited by authorities, arguing with the experience at prior demonstrations that participants would act against against the official restrictions. The ban was confirmed by Berlin's administrative court after the organizers of the protest had filed a suit against the initial prohibition. In spite of the ban, several hundred protesters gathered in different spots in Berlin e.g. on the 21st, 28th and 29th of August 2021.

The mentioned bans of large demonstrations of the 'Querdenken' movement as well as the general restrictions for demonstrations and public gatherings, limiting the number of participants to a maximum of twenty persons, decreased the momentum of the large, centralized protests focussing on the German legislative located in Berlin. However, the protests did not stop as a result, but rather changed their character: They became more decentralized and a shift to more informal forms of gatherings and protests could be observed: In several German cities and towns, people gathered for 'Mahnwachen' or 'Spaziergänge' (walks) to protest against the government's COVID-19 politics. Since going for a walk individually was a legal outdoor activity even under the strictest lock-down regulations, this type of action was chosen in order to circumvent the restrictions imposed on demonstrations. In effect, these walks ('Spaziergänge') often resulted in large gatherings of people moving through the streets together, often holding signs or chanting slogans against German politicians or the COVID-19 containment measures in general.

However, the calls to gather for walks in the public became more targeted and mobilized people to gather in front of the private homes of politicians, mostly those in charge for the COVID-19 measures on federal state level.

In December 2021, around thirty members of the far-right group 'Freie Sachsen' (Free Saxonians) gathered in front of the home of Saxony's Minister of Health, Petra Köpping, carrying torchlights. The incident received a lot of public attention, and was criticized in the media as well as by German politicians as attempts to intimidate politicians under the pretext of the protesters right to ‘freedom of speech’.

Earlier that year, the same group had gathered in front of Saxony’s prime minister, Michael Kretschmer, who is one of the most prominent targets of the radical movement against the COVID-19 containment measures. In December 2021, an investigative report by German journalists revealed that members of the Telegram chat group ‘Dresden Offlinevernetzung’ had talked about concrete preparations to assassinate Kretschmer. A few months later, by mid-April 2022, police foiled a plot to kidnap Germany’s Minister of Health Karl Lauterbach and to sabotage power facilities in order to create chaos which would ultimately allow to overthrow the government.

The messenger service Telegram played a key role in the mobilization of the movement. The service allows the creation of large groups and channels to disseminate information quickly to a large number of users. Unlike Facebook or Google, which hand over data to authorities at their request, Telegram does not cooperate with authorities.

This ensures a high degree of anonymity for its users which has proven beneficial for protest movements in authoritarian countries, This anonymity poses a barrier to political repression and criminal prosecution. This makes the service attractive not only for social movements, but also for criminal and terrorist activities, and also facilitates the spread of hate-speech and calls for violence like those mentioned above.

The rise of conspiracy theories in the context of the COVID-19 pandemic has thus not only intensified hate-speech against and the discrimination of certain communities in the form of antisemitism and racism - this process was accompanied by an increasingly toxic and potentially violent climate for politicians in Germany. This makes it once more clear that the digital sphere of social networks and messengers cannot be grasped without also considering its entanglement with and effects for the physical sphere of society. Both online and offline contexts influence and often reinforce each other, as is the case with the mobilization against the COVID-19 containment measures.

The processes illustrated above also point out another important aspect: Even though political activists from Germany's extremist far-right had been involved in the protests from the start, and non-governmental organizations had warned about the antidemocratic and violent potential of parts of the movement, the dangerous potential of this movement seems to have been underestimated by German authorities for a rather long time. Only after journalists had revealed the plans to murder Michael Kretschmer, and after several gatherings in front of private homes of politicians had already taken place, the issue seems to become more of a serious concern.

Thus the process points out the relevance of investigative work by critical journalists and non-governmental organizations to monitor the mobilization and networking strategies of the far-right spectre in Germany. However, while these actors often have to deal with large amount of data, their investigation often relies on manually conducted research and data evaluation. The potentials of Data Science methods, especially Machine Learning-based approaches for Natural Language Processing (NLP), are often not accessible for these actors, making it hard for them to process large amounts of data from the often very dynamic contexts of informal political networks.

- racism: especially Asian community made responsible for spreading the virus
- antisemitism: Jews identified as culprits behind the pandemic, using it as a means to obtain world dominance
- politicians: complicit with those who benefit from pandemic; following a secret plan; restriction of civil rights
- freedom of speech, criticizing the government are valuable and essential components of a democratic society
- but: often hateful tendencies and developments, both online and offline

- entanglement of digital and real world
- hate does not stay in the internet, also has real effects
- spread of misinformation, conspiracy theories: social media and messengers
- not only information, also mobilization

1.2 Objective

This thesis aims at exploring the potentials of a particular NLP technology, namely Named Entity Recognition (NER) for the investigation of the mobilization for protests against COVID-19 measures in a German-language Telegram channel. The overarching goal is to explore if a component for Named Entity Recognition could be a beneficial contribution to the Open Source project 'Teledash', to make NER available as a tool for journalists, NGOs, and researchers investigating the COVID-19 related protest movements and networks in Germany.

- low-resource approach of NER in German-language Telegram messages
- extraction of entities particularly relevant for mobilization / monitoring of potential targeting: PERSON, ORGANIZATION; DATE; TIME; LOCATION; ACTION as custom type
- explore potentials and limitations of existing models
- explore potentials of translation approach
- provide a low-resource tool for NGOs and researchersm make available as API; Project teledash

1.3 Thesis Structure

- Theory
- Implementation
- Conclusion and Discussion

Chapter 2

Theory

The first part of this chapter places Named Entity Recognition in its larger context of Natural Language Processing (NLP) in general, and of Information Extraction (IE) as a specific NLP task in particular. It provides a basic definition of of NER, lines out its historical evolvement, and discusses the challenges related to this task, both in general and with regard to the peculiarities of German language and to text data from social media.

The second part elaborates different approaches for NER, from heuristics-/rule-based methods over classical machine learning-based to deep learning-based approaches including transformers. It also discusses the limitations and potentials of the respective approaches.

2.1 Natural Language Processing

2.1.1 What is Natural Language Processing?

The field of Natural Language Processing combines approaches from computer science, linguistics, and artificial intelligence. Its main focus is to build systems and computational techniques for the automated processing and analysis of human language (cf. Vajjala et al. 2020, p. xvii). Since its evolvement in the 1950s, NLP has been primarily an area of interest for academic research, but lately, it is being used more and more for real-world tasks and business applications. (cf. Vajjala et al. 2020, p. xvii)

Core NLP applications comprise the use for email platforms (e.g. for spam filters), voice-based assistants, modern search engines, and machine translation services. (cf. Vajjala et al. 2020, p. 5). Furthermore, NLP is used for analyzing web and social media data (e.g. customer reviews), for spelling and grammar correction tools, and for a variety of domains including health care or finance. (cf. Vajjala et al. 2020, pp. 5–6).

Typical NLP tasks are for example language modeling, text classification, text summarization, information extraction, information retrieval, machine translation, or topic modeling (cf. Vajjala et al. 2020, pp. 6–7).

2.1.2 Challenges for NLP

Since NLP processes data representing human language, it is confronted with several challenges which stem from the characteristics of human language:

Ambiguity, understood as uncertainty or inexactness of meaning, is inherent to most human languages: Words or phrases can have multiple meanings, and usually it requires knowledge of the context in which they appear to resolve this ambiguity. Making computer-based systems take into account and 'understand' these contexts is one of the major challenges for NLP (cf. Vajjala et al. 2020, pp. 12–13)

Common knowledge is often an essential part in a conversation between humans: Facts are assumed to be known and thus not expressed explicitly in statements. Humans constantly use common knowledge to understand language. One of the challenges for NLP is thus how to encode this common knowledge for computational approaches (cf. Vajjala et al. 2020, pp. 13–14).

Creativity is another feature inherent to human language: Even though language is constituted via and essentially shaped by rules, it is far from being exclusively rule-driven. Rather, language is creative and heterogenuous; characterized by various styles, genres, and dialects. One important challenge for NLP is thus to enable machines to adequately handle this creativity. (cf. Vajjala et al. 2020, p. 14) Moreover, language changes and evolves over time, in interaction with social and cultural processes, as well as with the development of communication technology (one example being the emergence of hashtags in the context of social media). NLP applications developed using data from the past might thus not fit well for data consisting of newly evolving manifestations of human language.

Diversity across languages According to the database *Ethnologue*, there are currently over 7.000 languages actively spoken worldwide (cf. Ethnologue 2022). Even though some of these languages share the same family and thus have some similarities, a direct mapping between two languages is not possible. This makes it hard to apply an NLP solution from one language to another. Building language-agnostic solutions is a very complex matter, while building separate solutions for each language is labor- and time-intensive (cf. Vajjala et al. 2020, p. 14).

Imbalanced resources for different languages is yet another challenge for NLP, both in terms of research efforts and data. In fact, most NLP technologies are still designed for English or European languages (cf. Singh 2008). For these so-called high-resource languages, a large number of datasets and libraries exists while low-resource languages are usually significantly 'less studied, resource scarce, less computerized' (Magueresse, Carles, and Heetderks 2020). This imbalance is also reflected in the existing NER datasets.

2.2 Information Extraction

2.2.1 What is Information Extraction?

Information Extraction (IE) is one of the several NLP tasks mentioned above. The aim of IE is to extract information from unstructured or semistructured natural language texts (cf. Zong, Xia, and Zhang 2021, p. 227).

'Information' can refer to different things, depending on the domain and purpose of the task: The extracted instances can be (named) entities such as people, locations, places, or organizations, but also important events occuring in texts (cf. Vajjala et al. 2020, p. 161). Sometimes the focus is on temporal information such as time and date, or on domain-specific entities, e.g. biomedical instances like genes, proteins, or viruses.

The extracted parts can also be attributes related to those entities, or relationships between the detected entities. The kinds of texts which information is extracted from may be news texts from the web, social media posts, customer reviews, academic literature, and others. The aim of IE is to generate structured output from this unstructured or semistructured text data. Typical tasks of IE are named entity recognition, entity disambiguation, relation extraction, and event extraction (cf. (Zong, Xia, and Zhang 2021, p. 227)

2.2.2 Historical evolvement

The origins of IE dates back to the late 1970s, and its development was further accelerated in the late 1980s, when the US government financed activities to evaluate IE technology. Organized by the *Defense Advanced Research Projects Agency* (DARPA), the first *Message Understanding Conference* (MUC-1) took place in 1987. The conference provided standard datasets for international researchers to compete on. The MUC evaluation was held seven times between 1987 and 1997. The tasks included NER, coreference resolution, relation extraction, and slot filling, and mainly focused on text data from limited domains such as naval military intelligence, terrorist attacks, or aircraft crashes (cf. Zong, Xia, and Zhang 2021, p. 228).

By the end of the 1990s, the MUC was replaced by the *Automatic Content Extraction* (ACE) meeting which put its focus on extracting more fine-grained entities, as well as entity relations, and events. There was also a shift in the data to broader news data covering political and international events.

In 2009, ACE became a subtask of the *Text Analysis Conference Knowledge Base Population* (TAC KBP). This shifted the focus in data to open domain data (mainly from webpages), and the focus of tasks to entity attribute extraction and entity linking. Both MUC and ACE contributed to the development of several standard test datasets for the evaluation and development of IE approaches. (cf. Zong, Xia, and Zhang 2021, p. 229).

Regarding NER, another important event series was the *Conference on Computational Natural Language Learning* (CoNLL), which launched language-independent NER tasks in 2002 and 2003 (cf. Zong, Xia, and Zhang 2021, p. 229).

2.2.3 Classification and application of IE

Following Zong, Xia, and Zhang 2021, IE technology can be classified via the domain aspect (limited vs. open domain), the type of extracted information (entity recognition, relation extraction, event extraction), and implementation methods (rule-based, machine learning-based, or deep learning-based approaches) (cf. Zong, Xia, and Zhang 2021, p. 229).

Common examples of IE applications are news tracking, counter-terrorism, business and financial intelligence, the processing of biomedical data or scientific libraries. IE can also be utilized for applications building on entity-oriented search, for question-answering systems, or the task of text segmentation (cf. Aggarwal 2018).

2.3 Named Entity Recognition

2.3.1 What is Named Entity Recognition?

Named entity recognition is a subtask of IE. NER addresses the task of identifying named entities like person, location, organization, drug, time, clinical procedure, biological protein, etc. in text. It is often used as the first step in question answering, information retrieval, co-reference resolution, topic modeling, etc. (cf. Yadav and Bethard 2019). NER is thus a fundamental task in NLP. The entities which are to be identified can mainly be represented by the following categories:

- PERSON
- LOCATION
- ORGANIZATION
- MISC/OTH
- TIME
- DATE
- CURRENCY/QUANTITY, PERCENTAGE

While TIME, DATE, CURRENCY/QUANTITY, and PERCENTAGE can commonly be identified via regular expressions, instances of PERSON, LOCATION, ORGANIZATION, and MISC/OTH (miscellaneous/other entities) usually do not follow a consistent pattern and thus pose larger challenges for NER, which is why most of NER research focusses mainly on these entity types (cf. Zong, Xia, and Zhang 2021, p. 229).

Also it should be noted that TIME, DATE, CURRENCY/QUANTITY, and PERCENTAGE do not qualify as named entities in a narrow sense, but may be useful depending on the purpose of the application which NER is used for (cf. Aggarwal 2018, p. 386).

NER can be divided into the following two subtasks (cf.Zong, Xia, and Zhang 2021, p. 229):

- **1. Entity detection**, to detect whether a given string in a text document is an entity, including the detection of the entity's boundaries.
- **2. Entity classification**, to assign a specific category to the detected entity.

2.3.2 And how did it evolve?

The following section provides a brief history of the most relevant shared tasks for named entity recognition. The focus is on relevant cornerstones for NER in general, on German-language texts, and on data from social media.

MUC-6 The first NER task was introduced in 1996 at the Sixth Message Understanding Conference (MUC-6). The aim was to apply information extraction technologies for largely domain independent named entity recognition. Besides recognizing the entity types person, organization, and location, the task also involved numerical expressions including time, currency, and percentage (cf. Grishman and Sundheim 1996). The corpus for the task consisted of 318 annotated articles from the Wall Street Journal. The dataset is publicly available, but its use is copyright-restricted (cf. Chinchor and Sundheim 2003).

CoNLL 2002 and 2003 Further important corner stones for NER were the language-independent NER tasks at the Conference on Computational Natural Language Learning (CoNLL) for Spanish and Dutch in 2002 (cf. Tjong Kim Sang 2002), and for English and German in 2003 (cf. Tjong Kim Sang and De Meulder 2003).

The provided datasets consisted of newswire articles in the mentioned languages, and the task focuses on the entity types person (PER), location (LOC), organization (ORG), and miscellaneous (MISC).

Describing the task as language-independent in this context can be understood as developing systems which are not tuned specifically to the characteristics of one language, but can handle input data in different languages, meaning that the language is not known in advance.

The German dataset consists of 909 articles sampled from the newspaper Frankfurter Rundschau (cf. Tjong Kim Sang and De Meulder 2003).

The datasets are publicly available, but no licensing information is available. Given the fact that the corpus only contains newspaper articles, its applicability to other domains might be limited. Furthermore, the annotation of the English- and German-language datasets was carried out at the University of Antwerp (cf. Tjong Kim Sang and De Meulder 2003), thus most likely by non-native speakers, which might result in some inconsistencies.

GermEval 2014 In 2014, GermEval, a series of shared tasks focussing on Natural Language Processing for German-language, announced a task for Named Entity Recognition for German-language texts (cf. Benikova, Biemann, and Reznicek 2014).

The provided dataset was compiled from German Wikipedia as well as and online news. The named entity types of interest were person (PER), organization (ORG), location (LOC), and other (OTH), similar to CoNLL 2003 and 2004.

The dataset is available under a e CC-BY license which 'allows to distribute, alter and mix the data in any possible way and to use it for any purpose, including commercial ones.' (Benikova et al. 2014)

However, since the GermEval2014 dataset also includes nested and derived entities, it is not readily combinable with other datasets.

NER tasks for social media data In 2015, Baldwin et al. 2015 organized a shared task for Named Entity Recognition on social media data, namely tweets from Twitter in the context of the Workshop on Noisy User-generated Text (W-NUT). The task was designed to foster the development of NER systems capable of recognizing named entities in this specific domain which poses peculiar challenges to existing NER systems trained mainly on newswire text, since tweets, unlike edited newswire articles, contain a number of mis- and nonstandard spellings, abbreviations, incomplete sentences, and are often characterized by unreliable capitalization (cf. Baldwin et al. 2015).

As a result, existing NER tools mainly developed using newswire texts, perform rather poorly on this noisy, user-generated type of texts (cf. Baldwin et al. 2015).

Among the participating systems, the best F1 score achieved was 56.41, which is considerably lower than the best F1 score of 76.38 achieved at GermEval 2014 (cf. Benikova et al. 2014).

The data for the task was taken from the work by Ritter et al. 2011, who conducted an experimental study on named entity recognition in tweets, and used a random sample of 2.400 English-language tweets for this which were annotated with the following ten entity types: PERSON, GEO-LOCATION, COMPANY, PRODUCT, FACILITY, TV-SHOW, MOVIE, SPORTSTEAM, BAND, and OTHER (cf. Ritter et al. 2011).

The best participating system in this task achieved an F1 score of 0.66 for the ten types (cf. Ritter et al. 2011), thus outperforming the results later achieved in the shared task by Baldwin et al. 2015. Looking more detailed into the results of Ritter et al. 2011 for the standard entity types PERSON, LOCATION, and ORGANIZATION, it can be noted that the F1 score achieved for PERSON ranks highest with up to 0.83, followed by LOCATION, and then ORGANIZATION (0.74 and 0.66 for the best participating system).

2.3.3 What makes NER challenging?

As for NLP tasks in general, **ambiguity** if language is a core challenge (cf. Aggarwal 2018, p. 386). For example, the word 'Bill' might refer to a persons first name and thus indicate a named entity of type PERSON, but when located at the beginning of the sentence, it might also be a synonym for 'invoice'.

For resolving this ambiguity, the **context** of a given word is usually of crucial relevance (cf. Aggarwal 2018, p. 387). NER systems differ in terms of how they can or cannot take into account the context of a word. Moreover, the data itself might not offer enough context. While articles from newspapers usually are of sufficient length and contain sufficient contextual information, social media texts are often of shorter length which results in a lack of context.

Another important challenge are **abbreviations** which are used to refer to multiple instances of the same named entity (cf. Aggarwal 2018, p. 387). For example, the terms 'U.S.', 'USA' or 'United States of America' all refer to the same geopolitical entity. To overcome this challenge, usually entity disambiguation is an important step following named entity recognition.

The **creative and dynamic characteristics of human language** as described in 2.1.2 also pose challenges for the task of named entity recognition: Since **new entities** evolve over time, the set of know entities is never constant. While for 'some types of entities like locations, the names of entities evolve slowly, whereas for others like people and organizations, the incompleteness problem is very significant.' (Aggarwal 2018, p. 386).

Yet another challenge stems from capitalization conventions which vary across languages: While in English-language texts, capitalization usually indicates proper names and can thus be used as a feature for named entity recognition, in German-language texts all nouns are capitalized, making the feature less suitable (cf. Benikova, Biemann, and Reznicek 2014).

2.3.4 Making sense of noise: Specific challenges for NER in social media texts

The aforementioned challenges for NER also apply to texts from social media. However, named entity recognition for this genre can be regarded as even more challenging.

One reason for this stems from the challenges of **domain adaptation**: Since most of the existing general NER datasets consist of newswire texts, named entity recognition applications for other domains usually require the creation of custom annotated corpora. Moreover, social media text data is often more informal, noisy and shorter of length than newspaper articles, which aggravates domain adaptation for this genre and usually results in poor performance of existing NER systems when applied to texts from social media (cf. Ritter et al. 2011).

Moreover, since texts from social media are often shorter, they lack sufficient context for the determination of an entity's type (cf. Ritter et al. 2011).

The wide variety of capitalization styles are yet another challenge, like e.g. the use of only lowercase for German texts, or the use of uppercase only in some texts (cf. Ritter et al. 2011). As Ritter et al. 2011 points out, news-trained NER systems seem to rely heavily on capitalization which limits their applicability to user-generated social media texts. Figure 2.1 illustrates the noisy character of tweets as an example.

1	The Hobbit has FINALLY started filming! I cannot wait!
2	Yess! Yess! Its official Nintendo announced today that they Will release the Nintendo 3DS in north America march 27 for $250
3	Government confirms blast n nuclear plants n japan...don't knw wht s gona happen nw...

Figure 2.1: Examples of noisy text in tweets, taken from Ritter et al. 2011

Also, these texts usually contain more OOV (out of vocabulary) words than e.g. news articles. This is partly due to misspellings, but also stems from platform specific codes and styles, such as hashtags or lexical variations. For example, the authors collected more than fifty variations for the word 'tomorrow' from their Twitter corpus as shown in figure 2.2.

‘2m’, ‘2ma’, ‘2mar’, ‘2mara’, ‘2maro’, ‘2marrow’, ‘2mor’, ‘2mora’, ‘2moro’, ‘2morow’, ‘2morr’, ‘2morro’, ‘2morrow’, ‘2moz’, ‘2mr’, ‘2mro’, ‘2mrrw’, ‘2mrw’, ‘2mw’, ‘tmmrw’, ‘tmo’, ‘tmoro’, ‘tmorrow’, ‘tmoz’, ‘tmr’, ‘tmro’, ‘tmrow’, ‘tmrrow’, ‘tmrrw’, ‘tmrw’, ‘tmrww’, ‘tmw’, ‘tomaro’, ‘tomarow’, ‘tomarro’, ‘tomarrow’, ‘tomm’, ‘tommarow’, ‘tommarrow’, ‘tommoro’, ‘tommorow’, ‘tommorrow’, ‘tommorw’, ‘tommrow’, ‘tomo’, ‘tomolo’, ‘tomoro’, ‘tomorow’, ‘tomorro’, ‘tomorrw’, ‘tomoz’, ‘tomrw’, ‘tomz’

Figure 2.2: Lexical variations of the word ‘tomorrow’ in tweets, taken from Ritter et al. 2011

Last but not least, social media texts often combine different languages, making it harder for language-specific models to accurately determine named entities in them.

In 2015, Derczynski et al. 2015 investigated the applicability existing, by then state-of-the-art systems for named entity recognition in tweets as one example for data from microblogs. Their results indicate as well that existing machine-learning based approaches do not perform well on noisy user-generated data from microblogs. In line with Ritter et al. 2011, they identify unreliable capitalization, typographic errors, shortenings, and lack of context due to the shortness of the texts as main challenges in this specific domain. The authors suggest language detection, part-of-speech tagging, and normalization as pre-processing steps to encounter these challenges. Their results indicate however that e.g. normalization only leads to minimal increases, and in some cases even decreases the F1 score.

The authors furthermore point out that annotated datasets from the genre of microblogs are still really rare, and emphasize the need for creating larger human-annotated corpora of microblog content to allow better domain adaptation for named entity recognition systems (cf. Derczynski et al. 2015).

2.3.5 An overview of existing German-language datasets for NER

NER datasets

ConLL 2003 GermEval 2014 Smartdata corpus

first shared task on NER (Grishman and Sundheim, 1996) CoNLL 2002 (Tjong Kim Sang, 2002) and CoNLL 2003 (Tjong Kim Sang and De Meulder, 2003) were created from newswire articles in four different languages (Spanish, Dutch, English, and German) and focused on 4 entities - PER (person), LOC (location), ORG (organization) and MISC (miscellaneous including all other types of entities).

CoNLL 2003: licensing not clear, website not available; also: annotated by non-natve speakers, said to be somewhat inconsistent (cf. benikova 2014)

NER shared tasks have also been organized for a variety of other languages; German (Benikova et al., 2014) named entity types vary widely by source of dataset and language Benikova et al. (2014)'s data, which is based on German wikipedia and online news, has named entity types similar to that of CoNLL 2002 and 2003: PERson, ORGanization, LOCation and OTHer NER tasks have also been organized on social media data, e.g., Twitter, where the performance of classic NER systems degrades due to issues like variability in orthography and presence of grammatically incomplete sentences (Baldwin et al., 2015) Entity types on Twitter are also more variable (person, company, facility, band, sportsteam, movie, TV show, etc.) as they are based on user behavior on Twitter Though most named entity annotations are flat, some datasets include more complex structures; e.g. nested entities (Yadav and Bethard 2019)

2.3.6 Tagging scheme and entity types

B-I-O tagging scheme

- Researchers usually regard this task of supervised NER as a sequence labeling problem. The sequence labeling model first needs to determine the label set and the language granularity for labeling. BIO is a widely used label set in which "B" denotes the beginning of an entity, "I" indicates the middle or end of an entity, and "O" denotes the outside of any entity. Zong, Xia, and Zhang 2021, p. 231

For person, location, and organization names seven tags can be employed: PER-B, PER-I, LOC-B, LOC-I, ORG-B, ORG-1, and O. Among them, PER, LOC, and ORG denote person, location, and organization names, respectively. PER-B indicates the starting point of a person name, and PERI represents the middle or end part of a person name. LOC-B, LOC-I, ORG-B, and ORG-I have similar meanings.Zong, Xia, and Zhang 2021, pp. 231–232

Words and characters are usually the basic language units for label annotation.Zong, Xia, and Zhang 2021, p. 232

2.3.7 Evaluation of NER systems

Extrinsic evaluation

Intrinsic evaluation

'A named entity is correct only if it is an exact match of the corresponding entity in the data file.' Tjong Kim Sang 2002

Precision 'Precision is the percentage of named entities found by the learning system that are correct.' Tjong Kim Sang 2002

Recall 'Recall is the percentage of named entities present in the corpus that are found by the system.' Tjong Kim Sang 2002

F1 'The performance in this task is measured with $F_{\beta=1}$ rate which is equal to:

$$\frac{(\beta^2 + 1) * precision * recall}{(\beta^2 * precision + recall)}$$

with $\beta = 1$.

2.4 Different NER approaches

Early NER systems were based on handcrafted rules, lexicons, orthographic features and ontologies. followed by NER systems based on feature-engineering and machine learning (Nadeau and Sekine, 2007) Starting with Collobert et al. (2011), neural network NER systems with minimal feature engineering have become popular are appealing because they typically do not require domain specific resources like lexicons or ontologies, and are thus poised to be more domain independent. Various neural architectures have been proposed, mostly based on some form of recurrent neural networks (RNN) over characters, sub-words and/or word embeddings. Yadav and Bethard 2019

2.4.1 Rule-based approaches

Since both the internal structure and external context of person, location, and organization names have certain rules to follow, early research on NER mostly focused on rule-based methods, among which regular expression is commonly used. Compared to location and organization names, person names are much easier to identify for many languages (e.g., English, Chinese, and Japanese).

For example, in English, person names usually begin with capital letters and follow some kind of titles such as Mr. , Dr.,orProf.. Therefore, a regular expression title [capitalizedtoken+] can be designed to recognize such person names efficiently.Zong, Xia, and Zhang 2021, p. 230

- Nevertheless, even with a large-scale database of person, location, and organization names, rule-based NER methods still face many challenges. On the one hand, a phrase in different contexts may lead to different types of entities. Zong, Xia, and Zhang 2021, p. 231

- common words may also be a type of entityZong, Xia, and Zhang 2021, p. 231 e.g. Bill

- Furthermore, many entities are frequently abbreviated in the text. For example, United States of America is usually written as USA. This abbreviation can cause ambiguity in many cases. Zong, Xia, and Zhang 2021, p. 231 - More importantly, new named entities, especially for person and organization names, are constantly emerging, and the consistency becomes less reliable. These problems mean that rule-based methods enhanced with entity databases are difficult to handle and cannot obtain high recognition accuracy. Furthermore, the rulebased approach also faces the problem of system maintenance, as it requires the constant modification or addition of new rules that may conflict with existing ones. Therefore, the development of a NER model using an automatic learning framework is desirable. Zong, Xia, and Zhang 2021, p. 231

Each token in the text is converted into a set of features. These features are typically helpful properties of the token or their context for entity extraction. e.g. starts with capital letter help in defining various patterns on the left-hand side of the rule contextual pattern on the left-hand side of the rule is a combination of conditions corresponding to the features associated with a sequence of tokens a rule is fired if a sequence of tokens in the text matches this pattern action on the right-hand side could correspond to labeling that sequence as a named entity Aggarwal 2018

2.4.2 Knowledge-based approaches

Knowledge-based NER systems do not require annotated training data as they rely on lexicon resources and domain specific knowledge. These work well when the lexicon is exhaustive, but fail for entity classes not represented in the lexicon Precision is generally high for knowledge-based NER systems because of the lexicons, but recall is often low due to domain and language-specific rules and incomplete dictionaries Another drawback of knowledge based NER systems is the need of domain experts for constructing and maintaining the knowledge resources. Yadav and Bethard 2019

2.4.3 Machine-learning based approaches

Machine learning: "deals with the development of algorithms that can learn to perform tasks automatically based on a large number of examples, without requiring handcrafted rules." (cf. Vajjala et al. 2020, p. 14)

goal of ML: 'learn' to perform tasks based on examples (training data) without explicit instruction typically done by creating a numeric representation (features) of the training data and using this representation to learn the patterns in those examples (cf. Vajjala et al. 2020, p. 15)

ML algorithms can be grouped in three paradigms: supervised learning, unsupervised learning, reinforcement learning

supervised: learn the mapping function from input to output given a large number of examples in the form of input-output pairs input-output pairs = training data output = labels, or ground truth example: spam classification

supervised learning requires large number of labeled data (cf. Vajjala et al. 2020, p. 15)

Feature-engineered supervised systems

Given a collection of text data, suppose all person, location, and organization entities are manually labeled, as shown in the examples below. Supervised

NER systems attempt to design machine learning methods that learn automatic prediction models based on these correctly labeled training data.Zong, Xia, and Zhang 2021, p. 231

- Researchers usually regard this task of supervised NER as a sequence labeling problem.

Supervised machine learning models learn to make predictions by training on example inputs and their expected outputs, and can be used to replace human curated rules. Common ML systems for NER: Hidden Markov Models (HMM) Maximum Entropy Support Vector Machines (SVM) Conditional Random Fields (CRF) decision trees Yadav and Bethard 2019

Support Vector Machines - goal of any classification task: learn a decision boundary that acts as a separation between different categories of text - decision boundary can be linear or nonlinear (e.g. a circle) - SVM: can learn linear and nonlinear - SVM learns an optimal decision boundary so that the distance between points across classes is at its maximum Vajjala et al. 2020, p. 20

Hidden Markov Model HMM is a statistical model that assumes there is an underlying, observable process with hidden states that generates the data, ie we can only observe the data once it is generated - an HMM then tries to model the hidden states from this data - e.g. POS tagging: we assume that text is generated according to an underlying grammar, which is hidden underneath the text - hidden states are parts of speech that inherently define the structure of the sentence following the language grammar, but we only observe the words that are governed by these latent states - HMM also make the Markow assumption: each hidden state is dependent on the previous state(s) - human language is sequential which means that the current word depends on what occured before - this makes HMM a powerful tool for modeling textual data Vajjala et al. 2020, pp. 21–22

Conditional Random Field - another algorithm used for sequential data - performs a classification task on each element in the sequence - takes the sequential input and the context of tags into consideration Vajjala et al. 2020, p. 22

Unsupervised / bootstrapped approaches

Unsupervised learning: aim to find hidden patterns in given input data without any reference to output works with large collection of unlabeled data example: topic modeling, identity latent topics in large collection of texts (cf. Vajjala et al. 2020, p. 15)

common in real-world NLP: semi-supervised; small labeled dataset, large unlabeled dataset; learn from both sets (cf. Vajjala et al. 2020, p. 16)

reinforcement learning: absence of larger data colletions, trial and error principle, improvement via feedback (punishment or reward) (cf. Vajjala et al. 2020, p. 16)

Some of the earliest systems required very minimal training data. - labeled seeds, and 7 features including orthography (e.g., capitalization), context of the entity, words contained within named entities, etc. - shallow syntactic knowledge and inverse document frequency (IDF) - uses seeds to discover text having potential named entities, detects noun phrases and filters any with low IDF values, and feeds the filtered list to a classifier Yadav and Bethard 2019

2.4.4 Deep-learning based approaches / Neural Networks

Deep learning: "branch of machine learning that is based on artificial neural network architectures." (cf. Vajjala et al. 2020, p. 14)

NER systems have been studied and developed widely for decades but accurate systems using deep neural networks (NN) have only been introduced in the last few years (Yadav and Bethard 2019)

Feature-inferring neural network systems Collobert and Weston (2008) proposed one of the first neural network architectures for NER, with feature vectors constructed from orthographic features (e.g., capitalization of the first character), dictionaries and lexicons.Yadav and Bethard 2019

Later work replaced these manually constructed feature vectors with word embeddings (Collobert et al., 2011), which are representations of words in n-dimensional space, typically learned over large collections of unlabeled data through an unsupervised process such as the skip-gram model (Mikolov et al., 2013). Studies have shown the importance of such pre-trained word embeddings for neural network based NER systems (Habibi et al., 2017), and similarly for pre-trained character embeddings in character-based languages like Chinese Modern neural architectures for NER can be broadly classified into categories depending upon their representation of the words in a sentence. For example, representations may be based on words, characters, other sub-word units or any combination of these Yadav and Bethard 2019

Recurrent neural networks - language is inherently sequential - thus, a model which can progressively read an input text from one end to another can be very useful for language understanding - RNNs are designed to keep such a sequential processing and learning in mind - have neural units that are capable of remembering what they have processed so far - memory is temporal, information is stored and updated with every step as the RNN reads the next word as input Vajjala et al. 2020, p. 23

Long short-term memory - RNNs cannot remember longer contexts - do not perform well when input text is long - LSTM = solve this issue - concept: forget irrelevant context, only remember relevant context for task at hand - relieves the load of having to remember a lot of context for long inputsVajjala et al. 2020, p. 23

Convolutional neural networks - used heavily in computer vision tasks (e.g. image classification) - also used for NLP - each word in a sentence is represented as a word vector, all word vectors of sentence have the same size - resulting matrix representing the sentence can now be treated similar to an image by the CNN - advantage: CNN can look at a group of words together using a context window - CNN uses a collection of convolution and pooling layers to achieve a condensed representation of the text which is then fed as input to a fully connected layer to learn some NLP tasks like text classification Vajjala et al. 2020, p. 24

Transformers - latest entry in the league of deep learning models for NLP - have achieved state of the art in almost all major NLP tasks in recent years - model the textual context but not in a sequential manner - given a word in the input, it prefers to look at all the words around it (known as self-attention) and represent each word with respect to its context - e.g. bank different meaning depending on context (financial institute, furniture) - transformers can model such context and hence have been used heavily in NLP tasks due to this higher representation capacity as compared to other deep networks Vajjala et al. 2020, pp. 25–26

Transfer learning for downstream tasks with transformers - large transformers used for smaller downstream tasks via transfer learning - transfer learning = technique where the knowledge gained while solving one problem is applied to a different but related problem - transformers: pre-training (training of very large transformer models, unsupervised) to predict a part of a sentence given the rest of the content so it can encode the high-level nuances of the language in it - models are trained on more than 40GB of textual data, scraped from the whole internet - this model is then fine-tuned on NLP downstream tasks, e.g. NER Vajjala et al. 2020, p. 26

2.4.5 Summarization

Chapter 3

Implementation

3.1 Exploration of existing solutions

- spacy - Flair

- German - translated to English

explorative evaluation

3.2 Data

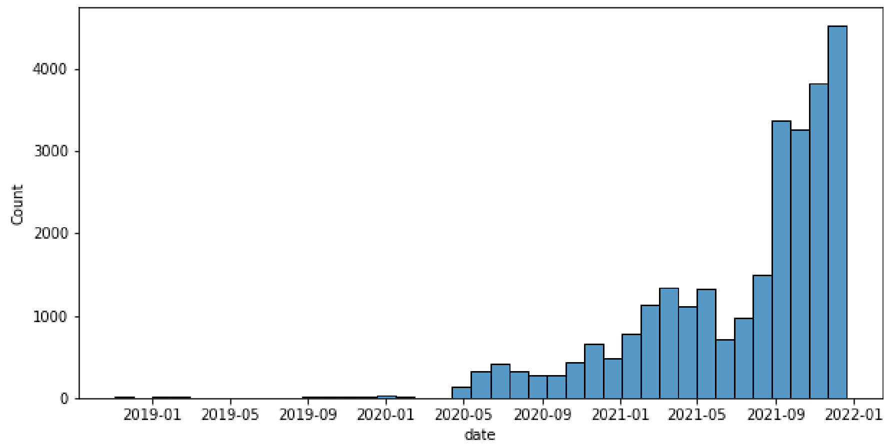
3.2.1 Domain-specific data

Telegram channel ‘Demotermine’: 52.900 subscribers by the beginning of June 2022,

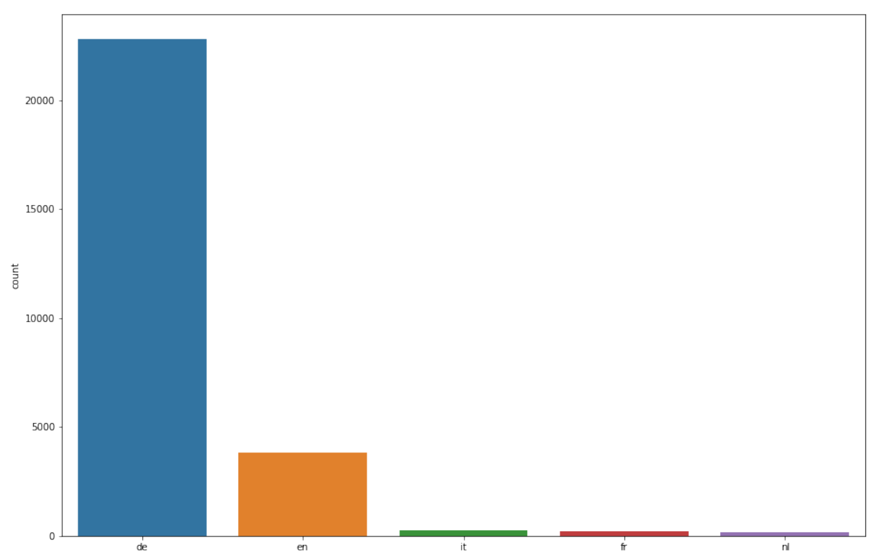
long channel name: ‘Demo-Termine & Kontakte, Gleichgesinnte treffen - Demokalender, Spaziergänge, Mahnwachen, Meditation *Rücktritt Bundesregierung!’

channel-info: “Wir Demokraten wollen miteinander in Kontakt bleiben @DemotermineChat und uns im Bedarfsfall möglichst schnell austauschen, Beweise in Wort, Bild und Ton sichern, damit sie nicht unterschlagen werden können @LivestreamsFuerDich! @Terminverteiler_Bot’

after removal of duplicates and NaN text:
date range: 2018-11-03 bis 2021-12-19



language: predominantly German (22803, ca. 83.064986%), followed by English (3.823, 13.926126%), French, Italian, Dutch (less than 1%)



qualitative exploration of data:

- origin of channel?

early messages (before pandemic):

- AfD, anti-immigration, Kandel, Gelbwesten - racist and antisemitic - against 'mass immigration' - focus on violence by immigrants - lying press - remembrance of German victims in WWII ('Bombenopfer'), self-victimization - pegida - landwirte gegen die politik der grünen - Julian Assange - against compulsary vaccinations (not COVID), Robert F. Kennedy jr. - GEGEN REPRESSION UND GROSSEN AUSTAUSCH - gegen USA (zB Ramstein airbase, drohnen, militärtransporte durch DE) - Dresden Gedenken, Rheinwiesenlager

"2019-09-01 11:01:38 Wir sehen uns auf der Straße ! Demos sind die ideale Gelegenheit , seinen Unmut gegen die herrschende Politik , die kein Interesse mehr am Volk hat (zum Ausdruck zu bringen . Hier lernt man vor allem Gleichgesinnte kennen , kann sich untereinander austauschen . Das Ziel bleibt , die Rechtsstaatlichkeit in der Bananenrepublik wiederherzustellen . Dazu gehört , dass in diesem Kanal keine Propaganda von Klima-Nazis (Buntfaschisten und steuerfinanzierten Gefährdern der freiheitlich-demokratischen Grundordnung (z.B. geteilt wird . Natürlich lehnen unsere Demoteilnehmer Nazi-Methoden der Terror-SA-Nachfolgeorganisation Antifa entschieden ab wie die durch Zwangsgebühren erpresste faschistische Propaganda von ARD und ZDF des Beitragsservice . Geteilt werden hier auch Demoankündigungen gegen die von Faschisten installierten Atomwaffen und durchgeführten Drohnenkriege von deutschem Boden aus . Grundsätzlich ist jede Form der Zionisierung , die die durch Migration massiv verstärkte Islamisierung zur Folge hat , abzulehnen . Hilfe an Hungernde hat grundsätzlich Vorrang gegenüber der faschistischen Deportation von wohlgenährten , jungen Männern nach und später vielleicht wieder Auschwitz . Menschen , die in Küstennähe ins Wasser gefallen sind , sind grundsätzlich zu retten und dort wieder an das nächste sichere Ufer an Land zu bringen , um ihr Leben und das von Nachahmern faschistischer Ideologien zu retten . Unsere Teilnehmer stehen zum Menschen- nach Artikel 5 im Grundgesetz . für das 5G-Netz und die des Regenwaldes zugunsten faschistischer , weltherrschaftlicher Ziele sind ebenso abzulehnen wie der faschistische Lobbyismus zum Impfzwang . Überall da , wo die feige , vermummte Antifa sein müsste , statt Faschisten die Rosetten zu lecken , sind wir da ! Soziale Ausgrenzung , Verbreitung von Hass und Hetze gegen Andersdenkende durch Stasi-Nazi-Faschisten nach Gleichmacher-Gleichschaltungs-Gleichschritt-Hitler-Vorbild liegt uns ebenso fern wie die Anwendung von Gewalt zugunsten der Diener des Faschismus in Parteien , Kirchen , Gewerkschaften , Stiftungen und EU-Reichs-Propagandamedien ARD und ZDF , die vor allem die stark wachsende Verarmung der Bevölkerung im reichen Fascho-Deutschland zur Folge hat , wogegen sich die auch Aktivisten wenden . Bei den Demos wollen wir überall und flächendeckend vor allem den Druck gegenüber der herrschenden Kaste aufbauen , den Menschen , die sich noch in der Wohlfühlblase und im Arschkriechermodus befinden zeigen , dass wir dennoch da sind ! Zu uns passende Demos bitte mitteilen . Bildet Fahrgemeinschaften , denn zusammen sind wir stark . Vernetzt euch , trefft einander . Das geht nirgendwo besser als auf der Straße . Dort erfährt man auch von weiteren (geschlossenen) und außerhalb der Demos ."

3.2.2 data cleaning

- total number of rows: 31430 - removal of duplicate texts - removal of rows not containing text

total number of messages (wo NaN / duplicates): 27.627 messages

3.2.3 text cleaning

some general and some Telegram- / channel-specific cleaning and preprocessing steps - removal of user mentions might refer to named entities, e.g. telegram channels run by / named after persons - but: explorations of models showed that this confuses models; also: mentions can easily be extracted from text via regex via the at the beginning, and are unambiguous, 100% accuracy can be achieved using regular expressions

- removal of urls: might contain strings containing named entities, confuse models - removal of sensitive information (bank account details in the form of IBANs), be able to make labeled data publicly available - normalization of unicode strings - removal of emojis - keep umlauts (German language) - removal of footer texts ('Raus auf die Straßen') - keep capitalization (relevant to identify common and proper nouns in German) - split up compound hashtags - tokenization, spacy tokenizer (german md model) - all tokens, not only text were kept (date, time consisting of numbers and punctuation, needed for the purpose of this NER application) - removal of duplicate whitespaces after joining tokens back together - removal of rows not containing values for cleaned text - 22.803 messages remaining

3.2.4 sample selection

- language detection with python package langdetect

- selected only German-language messages, German-language: 22.803 - selected only messages ¿ 100 characters including whitespaces (lower quartile: 83 chars) - 14.605 remaining

25%75

50%135

75%317

- identify similar messages via Levenshtein Distance - treshold: ¿= 0.8 similarity

- select random sample of 20% of remaining 11.978 messages - 2.395 messages for annotation remaining

Table 3.1: Annotation scheme: Entity types

Entity type	Description	Examples
PERSON	Named person or family, including artist names, names of fictional characters, etc.	
LOCATION	Countries, cities, states'; continents, districts, regions, streets and squares; shopping centers, parking lots, restaurants, 'Non-GPE locations, mountain ranges, bodies of water'; landmarks	
ORGANIZATION	Companies, agencies, institutions, newspapers, tv/radio channels, other media platforms; political parties; non-governmental organizations; political groups such as Querdenken or Antifa etc.	
DATE	Absolute or relative dates	23.09.2022
TIME	Specific times of day, time codes, periods of time; etc.	13:10
ACTION	Specific action or event mentioned in a text, e.g. demonstrations, pickets, motorcade; (protest) walks; etc. - must not be unique entity, plural allowed	
OTHER	Other entities, e.g. religions, nationalities, products, works of art, political spectres	

3.3 Annotation scheme

Based on Benikova, Biemann, and Reznicek 2014 and spacy

and on:

for the sake of simplicity: no nested entities, no derived entities

Only full noun phrases are considered as named entities. Pronouns and all other phrases can be ignored.

Names are denominators for a single unique object in the real world.

A named entity is a "real-world object" that's assigned a name – for example, a person, a country, a product or a book title. (spacy)

- NEs can consist of more than one token - Phrases only partly containing names (deutschlandweit), or adjectives referring to entities (euklidisch) can be ignored

combined NEs, e.g. Wien-Demo -¿ tagged separately if possible, otherwise tagged as the dominant entity type (here Demo), to avoid nested entities

hier wird klar wie der shift von protest forms is: Achtung Demonstration wird Abgesagt ! Wir haben als Orga Team beschlossen , das wir keine Demos mehr anmelden werden ! Wir sagen klar Nein zu den Auflagen die uns vorgegeben werden . Deshalb , neuer Termin , wir fahren nach Plauen ! Lasst uns die Freie Sachsen und den Bürgermeisterkandidaten Thomas Kaden unterstützen ! 11062021 Raus auf die Straße

labelstudio

IOB encoding: each word either begins, is inside, or is outside of a named entity

hashtags encoding events via location and date, e.g. #b20200809 or #le1311 are not to be annotated, should be extracted via regex and treated separately; unsure if they are NEs?

3.4 Annotated dataset

descriptive statistics access / licensing?

3.5 Additional resources

3.5.1 GermEval 2014 NER task dataset

3.5.2 DFKI Smartdata corpus

the DFKI SmartData Corpus is released as CC-BY 4.0.

3.5.3 XTREME dataset

3.6 Models

- spacy - hugging face

3.7 Pipeline

3.8 Translation approach

3.9 Evaluation

3.9.1 Evaluation metrics

3.9.2 Performance evaluation

3.10 Summarization

Chapter 4

Results

4.1 Spacy

4.2 Hugging Face

4.3 Translation approach

4.4 Summarization

Chapter 5

Conclusion

5.1 Summarization

5.2 Discussion

Bibliography

Aggarwal, Charu C. (2018). *Machine Learning for Text*. en. Cham: Springer International Publishing. ISBN: 978-3-319-73530-6 978-3-319-73531-3. DOI: 10.1007/978-3-319-73531-3. URL: http://link.springer.com/10.1007/978-3-319-73531-3 (visited on 05/14/2022).

Baldwin, Timothy et al. (July 2015). "Shared Tasks of the 2015 Workshop on Noisy User-generated Text: Twitter Lexical Normalization and Named Entity Recognition". In: *Proceedings of the Workshop on Noisy User-generated Text*. Beijing, China: Association for Computational Linguistics, pp. 126–135. DOI: 10.18653/v1/W15-4319. URL: https://aclanthology.org/W15-4319 (visited on 05/16/2022).

Benikova, Darina, Biemann, Chris, and Reznicek, Marc (May 2014). "NoSta-D Named Entity Annotation for German: Guidelines and Dataset". In: *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*. Reykjavik, Iceland: European Language Resources Association (ELRA), pp. 2524–2531. URL: http://www.lrec-conf.org/proceedings/lrec2014/pdf/276_Paper.pdf.

Benikova, Darina et al. (2014). "Germeval 2014 named entity recognition: Companion paper". In: *Proceedings of the KONVENS GermEval Shared Task on Named Entity Recognition, Hildesheim, Germany*, pp. 104–112.

Chinchor, Nancy and Sundheim, Beth (Aug. 2003). *Message Understanding Conference (MUC) 6*. Artwork Size: 10240 KB Pages: 10240 KB Type:

dataset. DOI: 10.35111/WBCC-Y063. URL: https://catalog.ldc.upenn.edu/LDC2003T13 (visited on 06/27/2022).

Derczynski, Leon et al. (Mar. 2015). "Analysis of named entity recognition and linking for tweets". en. In: *Information Processing & Management* 51.2, pp. 32–49. ISSN: 03064573. DOI: 10.1016/j.ipm.2014.10.006. URL: https://linkinghub.elsevier.com/retrieve/pii/S0306457314001034 (visited on 06/27/2022).

Ethnologue (May 2022). *How many languages are there in the world?* en. URL: https://www.ethnologue.com/guides/how-many-languages (visited on 06/19/2022).

Grishman, Ralph and Sundheim, Beth (1996). "Message Understanding Conference-6: A Brief History". In: *COLING 1996 Volume 1: The 16th International Conference on Computational Linguistics*. URL: https://aclanthology.org/C96-1079 (visited on 06/27/2022).

Magueresse, Alexandre, Carles, Vincent, and Heetderks, Evan (June 2020). *Low-resource Languages: A Review of Past Work and Future Challenges*. Number: arXiv:2006.07264 arXiv:2006.07264 [cs]. URL: http://arxiv.org/abs/2006.07264 (visited on 06/19/2022).

Mikolov, Tomas et al. (2013). *Efficient Estimation of Word Representations in Vector Space*. DOI: 10.48550/ARXIV.1301.3781. URL: https://arxiv.org/abs/1301.3781.

Ritter, Alan et al. (2011). "Named Entity Recognition in Tweets: An Experimental Study". In: *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. EMNLP '11. Edinburgh, United Kingdom: Association for Computational Linguistics, 1524–1534. ISBN: 9781937284114.

Schiersch, Martin et al. (2018). "A German Corpus for Fine-Grained Named Entity Recognition and Relation Extraction of Traffic and Industry Events". In: *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. Ed. by Nicoletta Calzolari (Conference chair) et al. Miyazaki, Japan: European Language Resources Association (ELRA). ISBN: 979-10-95546-00-9.

Singh, Anil Kumar (2008). "Natural Language Processing for Less Privileged Languages: Where do we come from? Where are we going?" In: *Proceedings of the IJCNLP-08 Workshop on NLP for Less Privileged Languages*. URL: `https://aclanthology.org/I08-3004` (visited on 06/19/2022).

Tjong Kim Sang, Erik F. (2002). "Introduction to the CoNLL-2002 Shared Task: Language-Independent Named Entity Recognition". In: *COLING-02: The 6th Conference on Natural Language Learning 2002 (CoNLL-2002)*. URL: `https://aclanthology.org/W02-2024`.

Tjong Kim Sang, Erik F. and De Meulder, Fien (2003). "Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition". In: *Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003*, pp. 142–147. URL: `https://aclanthology.org/W03-0419` (visited on 06/19/2022).

Vajjala, Sowmya et al. (2020). *Practical natural language processing: a comprehensive guide to building real-world NLP systems*. First edition. OCLC: on1125266646. Sebastopol, CA: O'Reilly Media. ISBN: 978-1-4920-5405-4.

Yadav, Vikas and Bethard, Steven (Oct. 2019). *A Survey on Recent Advances in Named Entity Recognition from Deep Learning models*. Tech. rep. arXiv:1910.11470. arXiv:1910.11470 [cs] type: article. arXiv. URL: `http://arxiv.org/abs/1910.11470` (visited on 05/14/2022).

Zong, Chengqing, Xia, Rui, and Zhang, Jiajun (2021). "Information Extraction". en. In: *Text Data Mining*. Singapore: Springer Singapore, pp. 227–283. ISBN: 9789811600999 9789811601002. DOI: `10.1007/978-981-16-0100-2_10`. URL: `https://link.springer.com/10.1007/978-981-16-0100-2_10` (visited on 05/14/2022).

Appendix

List of Figures

List of Tables

List of Acronyms